Sanath N Upadhyaya

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PROFESSIONAL SUMMARY

Software engineer with experience building backend components for large-scale distributed database systems and cloud infrastructure. Focused on scalability, fault tolerance, and observability, with hands-on experience in Kubernetes, CI/CD automation, and secure service design.

EDUCATION

Master of Science in Computer Science — Aug 2025 - Present
Northeastern University, Boston, MA — GPA: 4.0
Coursework: Foundations of Cryptography, Programming Design Paradigm, Reinforcement Learning, Algorithms, Database Systems
Graduate research: lattice-based cryptographic proof systems (zero-knowledge proofs), Khoury Cryptography & Privacy Group.

Bachelor of Engineering in Electronics and Communication Engineering — 2019 - 2023
BMS College of Engineering, Bengaluru, India — GPA: 9.15 / 10
Coursework: Operating Systems, Machine Learning, Internet of Things, Number Theory

WORK EXPERIENCE

Software Development Engineer Intern | Amazon Web Services, Palo Alto, CA — May 2026 - Aug 2026

- Built an automated integration-testing framework for Amazon Aurora (MySQL-compatible) from the ground up, generating and validating third-party workloads against distributed database clusters.
- Implemented differential-testing pipelines against reference databases to surface correctness discrepancies across query-execution paths.
- Designed agent-orchestration workflows to automate test generation, execution, and validation, cutting manual effort in large-scale compatibility testing.

Cloud Developer | Hewlett Packard Enterprise , Bengaluru, India — Aug 2023 - Aug 2025

- *Infrastructure Automation:* Led the development of Private Cloud AI Essentials (PCAI) on developer systems (XS) to enable faster iterative access to AI hardware, **reducing deployment time by 40%**.
- *Security Engineering:* Implemented Keycloak and OAuth2-based authentication pipelines; migrated from legacy LDAP to Greenlake IAM and Okta.
- *Zero-Trust Architecture:* Implemented zero-trust microservice authentication with Istio and SPIRE.
- *Prototyping & LLM Research:* Developed a multi-agent system generator using LangChain and CrewLLM as part of HPE's LLM Tool Mesh initiative.
- *CI/CD & Reliability*: Integrated backend services and automation into CI/CD pipelines (Jenkins, GitHub Actions), and added basic unit and integration tests using pytest and selenium to improve reliability of deployments.

Software Engineering Intern | Hewlett Packard Enterprise, Bengaluru, India — Jan 2023 - July 2023

- Implemented secure access for Ezmeral Runtime Engine using Keycloak and OAuth2-proxy; enabled OpenShift support for HPE Unified Analytics and optimized Prometheus performance.
- Automated Red Hat OpenShift cluster installation using Python and REST APIs, reducing deployment time by 30%.

PROJECTS

Akshara Mantapa: Infinite Kannada Text Library | Github **|** Web-Demo — **Rust, WebAssembly, SvelteKit**

- Designed and built a distributed text-generation and search system inspired by the Library of Babel, enabling deterministic generation and retrieval of massive synthetic text spaces.
- Implemented a Unicode-aware indexing pipeline using a 57k-symbol Kannada (Indic) grapheme cluster alphabet for correct segmentation and search
- Architected a Rust (Axum + WebAssembly) backend with a minimalist SvelteKit frontend for search and navigation

Post-Quantum & ZK Proof Benchmarking Framework | Github — **Rust, ml-kem, arkworks, Docker, Kubernetes**

- Designed HTTP microservices wrapping ml-kem (NIST FIPS 203) and arkworks R1CS Groth16 implementations.
- Measured ~17x throughput gap between lattice-based KEM and ZK proving across parameter sets.
- Containerized services with Kubernetes manifests for horizontal scaling experiments

End-to-End Encryption Engine with Double Ratchet | Github — **Java, Java Cryptography Extension (JCE),**

- Implemented Double Ratchet–based end-to-end encrypted asynchronous messaging in Java
- Achieved forward secrecy and post-compromise security using ECDH, HKDF, AES-GCM, and HMAC-SHA256

TECHNICAL SKILLS

Languages : Python, Go,Rust, C++, Java, Bash
Cloud Platforms: AWS (EC2, EKS, RDS, S3), GCP (GKE, Cloud Run), Azure
Distributed Systems and Infrastructure: PostgreSQL, MySQL, Redis, Apache Kafka, RabbitMQ
DevOps and Security: Jenkins, Github Actions , Prometheus, Ansible, Terraform, Keycloak
Data & AI : PyTorch,NumPy, Pandas,scikit-learn, OpenCV,Google-ADK, LangChain, CrewAI, MCP, MLX, RAG Pipelines
Certifications : Certified Kubernetes Administrator (**CKA**), Certified Kubernetes Security Specialist (**CKS**)